

Lecture 7 & 8: Assembly Language Programming (MCQs)

1. Assembly language is considered a _____ programming language.

- a) High-level b) Machine-independent c) Low-level d) Portable

• **Answer:** c) Low-level

2. The utility program that converts source code (.asm) into object code (.obj) is called _____.

- a) Linker b) Loader c) Editor d) Assembler

• **Answer:** d) Assembler

3. Which program combines separately assembled object modules into one executable (.exe) file?

- a) Assembler b) Linker c) Loader d) Editor

• **Answer:** b) Linker

4. .COM programs are restricted to a maximum size of _____.

- a) 1 MB b) 64 KB c) 32 KB d) Unlimited

• **Answer:** b) 64 KB

5. Which type of program contains separate segments for Code (CS), Data (DS), and Stack (SS)?

- a) .COM programs b) .EXE programs c) .OBJ files d) .LST files

• **Answer:** b) .EXE programs

6. In assembly language, the _____ register is typically used for looping.

- a) AX b) BX c) CX d) DX

• **Answer:** c) CX, *(Important: Repeated in Exam 2025 & Midterm 2024)*

7. Which register holds the address of the to be executed?

- a) Accumulator (AX) b) Stack Pointer (SP) c) Instruction Pointer (IP/PC) d) Data Register (DX)

• **Answer:** c) Instruction Pointer (IP/PC),

8. Which flag allows the programmer to step through execution one instruction at a time (Debugging)?

- a) ZF (Zero Flag) b) SF (Sign Flag) c) TF (Trap Flag) d) OF (Overflow Flag)

• **Answer:** c) TF (Trap Flag), *(Repeated in Exam 2023)*

9. Statements that tell the assembler to perform a specific action (like defining data) but generate are called _____.

- a) Instructions b) Operands c) Directives (Pseudo-ops) d) Comments

• **Answer:** c) Directives (Pseudo-ops),

10. Which of the following is a valid Assembly (generates machine code)?

a) SEGMENT b) MOV c) ENDP d) DB

• **Answer:** b) MOV

11. The directive DW stands for _____ and reserves _____ bits.

a) Define Word, 8 bits b) Define Word, 16 bits c) Define Double, 32 bits d) Data Write, 16 bits

• **Answer:** b) Define Word, 16 bits

****12. To reserve an uninitialized variable in memory, we use the symbol: ****

a) 0 b) * c)? d) null

• **Answer:** c)?

13. Which instruction is known to take (e.g., Instruction Op1, Op2, Op3)?

a) MOV b) ADD c) SHRD d) INC

• **Answer:** c) SHRD, (*Important: Repeated in Exam 2025 & Midterm 2024*)

14. In Assembly, a comment begins with a _____.

a) Slash (/) b) Hash # c) Semicolon (;) d) Star (*)

• **Answer:** c) Semicolon (;)

15. Which of the following is a valid identifier (Label) name?

a) _Start b) Start1 c) 1Start d) @Start

• **Answer:** c) 1Start, (*Identifiers cannot start with a digit - Midterm 2024 Question*)

16. The directive used to repeat a value multiple times (e.g., reserving an array) is _____.

a) REP b) LOOP c) DUP d) COPY

• **Answer:** c) DUP

17. MOV AX, [BX] is an instruction that takes _____ operand(s).

a) No b) One c) Two d) Three

• **Answer:** c) Two

18. .COM programs must start at offset _____, so they require ORG 100H.

a) 000H b) 100H c) 010H d) 200H

• **Answer:** b) 100H,

19. Which directive is used to mark the ?

a) ENDS b) END c) ENDP d) STOP

• **Answer:** c) ENDP

20. The ASSUME directive is used to _____. a) Initialize values in registers b) Tell the assembler which segment register belongs to which logical segment c) End the program d) Perform arithmetic operations

• **Answer:** b) Tell the assembler which segment register belongs to which logical segment

21. Which register is primarily used as the for arithmetic operations? a) AX b) BX c) CX d) SI

• **Answer:** a) AX

22. RET (Return) is an example of an instruction that has _____ operands. a) Zero b) One c) Two d) Three

• **Answer:** a) Zero

23. The directive DB allocates memory in chunks of _____. a) 1 Byte (8 bits) b) 1 Word (16 bits) c) 1 Double Word (32 bits) d) 10 Bytes

• **Answer:** a) 1 Byte (8 bits)

24. In the instruction ADD AX, BX, the result is stored in _____. a) BX b) AX c) Both AX and BX d) Memory

• **Answer:** b) AX (Destination)

25. Which of the following is a "Reserved Word" in Assembly and cannot be used as a label? a) COUNT b) TOTAL c) MOV d) MY_DATA

• **Answer:** c) MOV, (Exam 2025 Question)

26. The INC instruction (e.g., INC AX) takes _____ operand(s). a) 0 b) 1 c) 2 d) 3

• **Answer:** b) 1

27. To define a "Stack Segment", the combine type in the SEGMENT directive should be _____. a) PUBLIC b) COMMON c) STACK d) NONE

• **Answer:** c) STACK

28. Which file extension represents the "List file" generated by the assembler? a) .EXE b) .OBJ c) .LST d) .ASM

• **Answer:** c) .LST

29. SEGMENT and ENDS directives are used to define a _____. a) Procedure b) Macro c) Segment (Logical memory area) d) Variable

• **Answer:** c) Segment

30. Which flag is set to 1 if the result of an arithmetic operation is negative? a) ZF b) SF c) OF d) CF

• **Answer:** b) SF (Sign Flag)